THE INTEGRAL CECH CHAIN COMPLEX OF A COVERING FAMILY IN A CATEGORY IS EXACT IN POSITIVE DEGREES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed, and the proof is complete in the source, with no omitted verification.

Editorial note. The leading context span is the author's own Lemma lemma-cech-map-into (tag 03AS) of the same section, printed because the counted statement refers to the chain complex Z_{U,bullet} defined there, including its signs (-1)^j and the placement of the last nonzero term in degree 0. That span defines the label lemma-cech-map-into, which is therefore not listed among the external references; its own proof is not printed because the counted proof does not use it.

Editorial note. Because the counted statement and its printed context are self-contained with respect to the chapter, the card declares no external references at all; the two items of the standard prerequisite list are background notions, not cited labels.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex): a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros. The native class line, the xr-hyper externaldocument block and the reference/slogan/history display environments are omitted because the assembler supplies a plain amsart wrapper and the excerpt uses none of those environments; no macro was invented or redefined.

Allowed standard prerequisites (editorial).

- Free abelian presheaves represented by objects of the category, and the adjunction Hom(Z_X, F) = F(X). Modules on Sites, Stacks tag 03A8 (sites-modules-lemma-obvious-adjointness), used by the printed context lemma for the identification recorded at its proof line 1021. Fixes the meaning of the terms Z_{U_{i0} times_U ... times_U U_{ip}} out of which the chain complex of the statement is built; the exactness argument itself only uses free abelian groups on sets.
- Abelian presheaves on a category, morphism sets and fibre products in an arbitrary category. the standing conventions of the chapter, restated inside the statement at source lines 1026-1029. The statement only assumes that the fibre products U_{i0} times_U ... times_U U_{ip} exist in the category C; the proof works with the sets Mor_C(V, U_{i0} times_U ... times_U U_{ip}) and never needs a site structure.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be a family of morphisms with fixed target such that all fibre products $U_{i_0} \times_U \ldots \times_U U_{i_p}$ exist in $\mathcal{C}$. Consider*

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the chain complex $\mathbf{Z}_{\mathcal{U},\bullet}$ *of abelian presheaves*

$$\ldots \to \bigoplus_{i_0 i_1 i_2} \mathbf{Z}_{U_{i_0} \times_U U_{i_1} \times_U U_{i_2}} \to \bigoplus_{i_0 i_1} \mathbf{Z}_{U_{i_0} \times_U U_{i_1}} \to \bigoplus_{i_0} \mathbf{Z}_{U_{i_0}} \to 0 \to \ldots$$

where the last nonzero term is placed in degree 0 *and where the map*

$$\mathbf{Z}_{U_{i_0} \times_U \ldots \times_u U_{i_{p+1}}} \longrightarrow \mathbf{Z}_{U_{i_0} \times_U \ldots \widehat{U_{i_j}} \ldots \times_U U_{i_{p+1}}}$$

is given by $(-1)^j$ *times the canonical map. Then there is an isomorphism*

$$\mathrm{Hom}_{PAb(\mathcal{C})}(\mathbf{Z}_{\mathcal{U},\bullet}, \mathcal{F}) = \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$$

functorial in $\mathcal{F} \in \mathrm{Ob}(PAb(\mathcal{C}))$.

Lemma 0.2. *Let* $\mathcal{C}$ *be a category. Let* $\mathcal{U} = \{f_i : U_i \to U\}_{i \in I}$ *be a family of morphisms with fixed target such that all fibre products* $U_{i_0} \times_U \ldots \times_U U_{i_p}$ *exist in* $\mathcal{C}$*. The chain complex* $\mathbf{Z}_{\mathcal{U},\bullet}$ *of presheaves of Lemma 0.1 above is exact in positive degrees, i.e., the homology presheaves* $H_i(\mathbf{Z}_{\mathcal{U},\bullet})$ *are zero for* $i > 0$.

Proof. Let V be an object of $\mathcal{C}$. We have to show that the chain complex of abelian groups $\mathbf{Z}_{\mathcal{U},\bullet}(V)$ is exact in degrees > 0. This is the complex

$$\begin{array}{c} \ldots \\ \downarrow \\ \bigoplus_{i_0 i_1 i_2} \mathbf{Z}[\mathrm{Mor}_{\mathcal{C}}(V, U_{i_0} \times_U U_{i_1} \times_U U_{i_2})] \\ \downarrow \\ \bigoplus_{i_0 i_1} \mathbf{Z}[\mathrm{Mor}_{\mathcal{C}}(V, U_{i_0} \times_U U_{i_1})] \\ \downarrow \\ \bigoplus_{i_0} \mathbf{Z}[\mathrm{Mor}_{\mathcal{C}}(V, U_{i_0})] \\ \downarrow \\ 0 \end{array}$$

For any morphism $\varphi : V \to U$ denote $\mathrm{Mor}_\varphi(V, U_i) = \{\varphi_i : V \to U_i \mid f_i \circ \varphi_i = \varphi\}$. We will use a similar notation for $\mathrm{Mor}_\varphi(V, U_{i_0} \times_U \ldots \times_U U_{i_p})$. Note that composing with the various projection maps between the fibred products $U_{i_0} \times_U \ldots \times_U U_{i_p}$ preserves these morphism sets. Hence we see that the complex above is the same as the complex

$$\begin{array}{c} \ldots \\ \downarrow \\ \bigoplus_\varphi \bigoplus_{i_0 i_1 i_2} \mathbf{Z}[\mathrm{Mor}_\varphi(V, U_{i_0} \times_U U_{i_1} \times_U U_{i_2})] \\ \downarrow \\ \bigoplus_\varphi \bigoplus_{i_0 i_1} \mathbf{Z}[\mathrm{Mor}_\varphi(V, U_{i_0} \times_U U_{i_1})] \\ \downarrow \\ \bigoplus_\varphi \bigoplus_{i_0} \mathbf{Z}[\mathrm{Mor}_\varphi(V, U_{i_0})] \\ \downarrow \\ 0 \end{array}$$

Next, we make the remark that we have

$$\mathrm{Mor}_\varphi(V, U_{i_0} \times_U \ldots \times_U U_{i_p}) = \mathrm{Mor}_\varphi(V, U_{i_0}) \times \ldots \times \mathrm{Mor}_\varphi(V, U_{i_p})$$

Using this and the fact that $\mathbf{Z}[A] \oplus \mathbf{Z}[B] = \mathbf{Z}[A \amalg B]$ we see that the complex becomes

$$\begin{array}{c} \vdots \\ \downarrow \\ \bigoplus_\varphi \mathbf{Z}\left[\coprod_{i_0 i_1 i_2} \mathrm{Mor}_\varphi(V, U_{i_0}) \times \mathrm{Mor}_\varphi(V, U_{i_1}) \times \mathrm{Mor}_\varphi(V, U_{i_2})\right] \\ \downarrow \\ \bigoplus_\varphi \mathbf{Z}\left[\coprod_{i_0 i_1} \mathrm{Mor}_\varphi(V, U_{i_0}) \times \mathrm{Mor}_\varphi(V, U_{i_1})\right] \\ \downarrow \\ \bigoplus_\varphi \mathbf{Z}\left[\coprod_{i_0} \mathrm{Mor}_\varphi(V, U_{i_0})\right] \\ \downarrow \\ 0 \end{array}$$

Finally, on setting $S_\varphi = \coprod_{i \in I} \mathrm{Mor}_\varphi(V, U_i)$ we see that we get

$$\bigoplus\nolimits_\varphi (\ldots \to \mathbf{Z}[S_\varphi \times S_\varphi \times S_\varphi] \to \mathbf{Z}[S_\varphi \times S_\varphi] \to \mathbf{Z}[S_\varphi] \to 0 \to \ldots)$$

Thus we have simplified our task. Namely, it suffices to show that for any nonempty set S the (extended) complex of free abelian groups

$$\ldots \to \mathbf{Z}[S \times S \times S] \to \mathbf{Z}[S \times S] \to \mathbf{Z}[S] \xrightarrow{\Sigma} \mathbf{Z} \to 0 \to \ldots$$

is exact in all degrees. To see this fix an element $s \in S$, and use the homotopy

$$n_{(s_0, \ldots, s_p)} \longmapsto n_{(s, s_0, \ldots, s_p)}$$

with obvious notations. □